

Manual

Energy Management Key Performance Indicators EMG-KBW 8.1

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1 Key Figure Overview (Relating to Energy)

Summary

Possible fields of application

The "key figures" (relating to energy) function package allows for consolidated evaluations to be made of the data (quantities, durations, consumption values) recorded in MDE and EMG over different periods and relating to defined key figures.

Implementation notes

The function package is used if:

- You would like to monitor and assess the performance of individual machines, machine groups or departments on the basis key performance indicators relating to energy.
- You would like to assign consumption counters to individual machines for key figure calculation.

Integration

The times and quantities recorded in BDE or MDE as well as the consumption values entered in EMG are used for data representation.

Features

- Efficiency report (energy)
 - Efficiency report relating to quantities and time for all machines over individual periods and shifts
 - Recorded consumption values of the counters assigned to machines
 - Tabular evaluation of key performance indicators, specific energy consumption, energy consumption per machine hour and energy consumption per production hour.
- Assignment of counters to machines
 - Assignment of machines to counters as basis for the consideration of key performance indicators.

2 Consumption/Energy Monitor

Overview

Menu	Production Facility Management → Resource analysis → Energy monitor
Transaction code	conmon
Function authorization	conmon

This document provides a description of the "Consumption/ energy monitor" application in the Manufacturing Operation Center (MOC).

Purpose

The energy monitor lists the energy counters (resources). The energy monitor shows the current status for these resources, allowing a comparison with comparative resources. This makes it possible to display an energy balance. The system always shows the counted quantities since the last inventory, i.e. since the counter was reset.

You can also display the balance in a graphic. The same applies to the counter hierarchy defined by the BOM including the meter readings of the selected resource.

When you select data, you can use the defined counter hierarchies in tree-like structure to filter the resources in the table multiple times.

Integration

This application is connected with the following applications:

- Consumption analysis to chronologically distribute consumption values
- Consumption statement to analyze the totals and to reset data.
- You can also visualize the monitor values in the graphic machinery due to the integration of the counters (resources).

Selection criteria

Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can configure individual resource types for the other resources. Predefined resource types include:

DNC NC/DNC program

DOC	Document
ENT	Removal device
MNR	Workplace/Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

Resource from ... to ...

This selection criterion refers to the resource. You can also run a search using wildcards (placeholders * and ?).

Designation (name)

Name of the resource.

Cost center

Cost center of the resource.

Responsibility area

Responsibility area to which the resource is assigned.

Resource family

The resource family to which the resource is assigned.

Storage location

Regular storage location of the resource.

User fields

MD user fields 1- 6 of the resource.

Reference

The resource's internal ID.

Field descriptions*"General" category***Resource type**

Resource type of the resource. By default, the HYDRA system includes some default resource types. You can configure additional resource types in HYDRA.

Resource

Resource as a reference point for comparisons.

Designation (name)

Name of the resource.

Resource status

Resource status.

Cost center

Cost center

*"Resource" category***Current value**

Current value

Quantity unit (P)

Quantity unit (P)

Absolute value limit (only available if the EMG-KBW license has been purchased)

Absolute value limit

Reset time

Date on which the resource counters were reset.

Status time

Status time

"Key figures" category (only available if the EMG-KBW license has been purchased)**Energy consumption per hour**

Energy consumption per hour

Key figure 2 to 10

You can show up to nine additional key figures. Use the formula management to define the key figures (KPIs). You can configure the formulas eovf2 to eovf10.

*"Comparison resource 1/2" category***Resource 1 / Resource 2**

Resource whose values are to be compared with those of the reference point.

Current value

Absolute difference between the values of the reference resource and the comparison resource.

Difference, in percent

Relative difference between the reference resource values and the comparison resource values.

Unit

Unit

Resource type 1 / Resource type 2

Resource type of the comparison resource.

Reset time

Date on which the counters of the comparison resource were reset.

Status time

Status time

Setting a reset time

The reset date depends on the entered records.

If you select a reset date that coincides with a resource record period, the system sets the reset time to the start time of this resource record.



If you select a reset date that does not coincide with a resource record period, the system uses the start time of the next resource record as the reset time.

The system uses the reset time you enter if:

- you select a reset date that does not coincide with a resource record period and
- there is no record that starts at a later point in time.

3 Counter to Machine Assignment

Summary

Menu	Master data → Resources → Assignment of counter to machine
Transaction code	emgmas
Function authorization	emgmas

This document describes the "counter to machine assignment" application within the Manufacturing Operation Center (MOC).

Utilization

This application enables the assignment of one or several energy meter resources to every workplace.

This assignment is considered for the calculation of key figures within the efficiency report (relating to energy). Key performance indicators are calculated for every workplace/counter assignment.

Please note

It is not allowed to assign two or more counters of the same resource type and the same resource family to one workplace. However, several meter resources of different types or different families may be assigned.

One counter may be assigned to multiple workplaces.



The family structure of (meter) resources is to be focused in particular, due to the variety of assignment options and the resulting evaluation options within the efficiency report (energy).

Toolbar



Add

Opens the dialog for adding an assignment.



Delete

Deletes one or several assignments.

Selection criteria

The following selection criteria are available in the application:

Workplace

Workplace/machine that is to be assigned a counter resource.
Wildcards (placeholders *) can be used.

Energy meter

Selects the assigned energy counters. Wildcards (placeholders *) can be used.

Field descriptions

Workplace category

Resource type

Resource type of the workplace

Workplace

Number of the workplace

Assigned or energy meter category

Resource type

Resource type of the energy meter resource.

Resource family

Resource family which the energy meter resource belongs to.

Resource

The number of the energy meter resource

Last modified on category

Editor

The last person to edit this data record

Modified on

Date on which this data record was last modified.

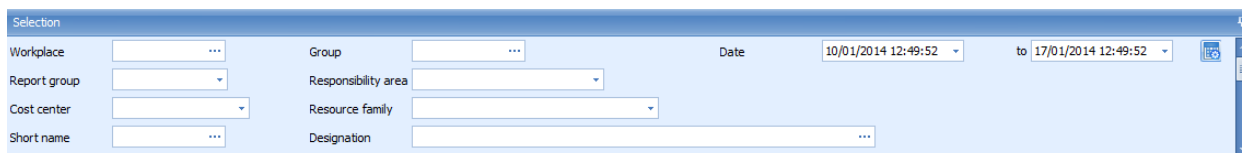
4 Efficiency Report (Relating to Energy)

Summary

Menu	Production facility management → Key performance indicators → Efficiency report (energy)
Transaction code	effremg
Function authorization	effremg

The analysis concerns workplace/ machine-related performance data for a certain period of time and a certain number of workplaces. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

Selection criteria



The following selection criteria are available in the application:

Workplace

This selection criterion references the workplace in the machine or workplace master data. Wildcards (placeholders *) can be used.

Group

This selection criterion references the group in the machine or workplace master data. All machines or workplaces are displayed that are assigned to the selected group. Wildcards can be used.

Date/time from ... to ...

Restricts the period to be evaluated by filling out the from/to fields

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Responsibility area

This selection criterion refers to the responsibility area within the workplace/machine master. Please respect that only machines are displayed, for which the user is authorized by the corresponding responsibility areas.

Cost center

This selection criterion refers to the cost center stored in the machine or workplace master. All machines or workplaces are displayed that are assigned to the selected cost center. Wildcards can be used.

Resource family

This selection criterion refers to the family of (counter) resources assigned to the workplaces/machines.

Short name

This selection criterion refers to the short name of the machines within master data. All of the machines or workplaces are displayed that match the string that was entered. Wildcards can be used.

Designation

This field refers to the designation of machines and workplaces within the machine master data. Only those machines are displayed that are identical to the string that was entered. Wildcards (*) can be used in this field.

Efficiency report detail application

Workplace category

The following workplace/ machine-related master data are available:

- Workplace
- Short name
- Designation
- Group
- Cost center

Duration category

- Production = RPA11
- Downtime = RPA01 + RPA02 + RPA03 + RPA04 + RPA05 + RPA06 + RPA07 + RPA08 + RPA09 + RPA10
- Total = production + downtime

Primary quantity, secondary quantity, tertiary quantity, basic quantity category

Workplace/ machine-related quantities recorded in the corresponding quantity types

- Yield
- Scrap
- Rework
- Open quantity

or the relevant quantity units (if relevant in the customer system).

Cycles category

- Number of posted cycles

"Energy meter" category

These master data are available for the energy meter resource:

- Resource
- Designation
- Resource family
- Consumption
- Unit
- Status time

Key figures category

- Specific energy consumption

These key figures represent the specific energy demand relating to the production quantity.

Specific energy consumption = energy consumption / yield (P)

- Energy consumption per machine hour

These key figures represent energy demand relating to machine hours

Energy consumption per machine hour = total energy consumption / total machine hours

- Energy consumption per production hour

These key figures represent energy demand relating to production hours

Energy consumption per production hour = total energy consumption per machine hour for RPA 11

Key figure 4 to..10

Up to seven additional key figures may be shown. Key figures are defined within formula management. The formulas erpf4 to erpf10 are defined as part of customizing the system.

"Resource performance accounts" category

- RPA 1-10 and RPA12